FINITE-TYPE APPROXIMATION PRESERVING A PRESCRIBED INFINITESIMAL OBJECT AND ITS ASSOCIATED GRADED RING

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Editorial note. Both the G-ring hypothesis at the image point and limit preservation on objects remain explicit; no unrestricted algebraization theorem is asserted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. G-rings, limit-preserving fibre categories, Artin approximation in its stated G-ring variant, complete local presentations, Artin–Rees and the standard stability criterion for associated graded complexes. The simultaneous choice of equations, relations and object is the retained proof obligation.

Source reference. Original lemma 01TA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2367–2367. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00QN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 973–973. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06GZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8303–8303. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 07VD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10003–10003. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 07PV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9810–9810. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 07QZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9849–9849. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:artin.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 07VF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10005–10005. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00CR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 471–471. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Let x be an object of $\mathcal{X}$ lying over $\mathrm{Spec}(R)$ where R is a Noetherian complete local ring with residue field k of finite type over S. Let $s \in S$ be the image of $\mathrm{Spec}(k) \to S$. Assume that (a) $\mathcal{O}_{S,s}$ is a G-ring and (b) p is limit preserving on objects. Then for every integer $N \geq 1$ there exist*

1. *a finite type S-algebra A,*
2. *a maximal ideal $\mathfrak{m}_A \subset A$,*
3. *an object x_A of $\mathcal{X}$ over $\mathrm{Spec}(A)$,*
4. *an S-isomorphism $R/\mathfrak{m}_R^N \cong A/\mathfrak{m}_A^N$,*
5. *an isomorphism $x|_{\mathrm{Spec}(R/\mathfrak{m}_R^N)} \cong x_A|_{\mathrm{Spec}(A/\mathfrak{m}_A^N)}$ compatible with (4), and*
6. *an isomorphism $Gr_{\mathfrak{m}_R}(R) \cong Gr_{\mathfrak{m}_A}(A)$ of graded k-algebras.*

Proof. Choose an affine open $\mathrm{Spec}(\Lambda) \subset S$ such that k is a finite Λ-algebra, see Morphisms, Lemma 01TA. We may and do replace S by $\mathrm{Spec}(\Lambda)$.

We may write R as a directed colimit $R = \mathrm{colim}\, C_j$ where each C_j is a finite type Λ-algebra (see Algebra, Lemma 00QN). By assumption (b) the object x is isomorphic to the restriction of an object over one of the C_j. Hence we may choose a finite type Λ-algebra C, a Λ-algebra map $C \to R$, and an object x_C of $\mathcal{X}$ over $\mathrm{Spec}(C)$ such that $x = x_C|_{\mathrm{Spec}(R)}$. The choice of C is a bookkeeping device and could be avoided. For later use, let us write $C = \Lambda[y_1, \ldots, y_u]/(f_1, \ldots, f_v)$ and we denote $\overline{a}_i \in R$ the image of y_i under the map $C \to R$. Set $\mathfrak{m}_C = C \cap \mathfrak{m}_R$.

Choose a Λ-algebra surjection $\Lambda[x_1, \ldots, x_s] \to k$ and denote $\mathfrak{m}'$ the kernel. By the universal property of polynomial rings we may lift this to a Λ-algebra map $\Lambda[x_1, \ldots, x_s] \to R$. We add some variables (i.e., we increase s a bit) mapping to generators of $\mathfrak{m}_R$. Having done this we see that $\Lambda[x_1, \ldots, x_s] \to R/\mathfrak{m}_R^2$ is surjective. Then we see that

$$P = \Lambda[x_1, \ldots, x_s]^\wedge_{\mathfrak{m}'} \longrightarrow R \tag{0.1.1}$$

is a surjective map of Noetherian complete local rings, see for example Formal Deformation Theory, Lemma 06GZ.

Choose lifts $a_i \in P$ of $\overline{a}_i$ we found above. Choose generators $b_1, \ldots, b_r \in P$ for the kernel of (0.1.1). Choose $c_{ji} \in P$ such that

$$f_j(a_1, \ldots, a_u) = \sum c_{ji} b_i$$

in P which is possible by the choices made so far. Choose generators

$$k_1, \ldots, k_t \in \mathrm{Ker}(P^{\oplus r} \xrightarrow{(b_1, \ldots, b_r)} P)$$

and write $k_i = (k_{i1}, \ldots, k_{ir})$ and $K = (k_{ij})$ so that

$$P^{\oplus t} \xrightarrow{K} P^{\oplus r} \xrightarrow{(b_1, \ldots, b_r)} P \to R \to 0$$

is an exact sequence of P-modules. In particular we have $\sum k_{ij}b_j = 0$. After possibly increasing N we may assume $N-1$ works in the Artin-Rees lemma for the first two maps of this exact sequence (see More on Algebra, Section 07VD for terminology).

By assumption $\mathcal{O}_{S,s} = \Lambda_{\Lambda\cap\mathfrak{m}'}$ is a G-ring. Hence by More on Algebra, Proposition 07PV the ring $\Lambda[x_1,\ldots,x_s]_{\mathfrak{m}'}$ is a G-ring. Hence by Smoothing Ring Maps, Theorem 07QZ there exist an étale ring map

$$\Lambda[x_1,\ldots,x_s]_{\mathfrak{m}'} \to B,$$

a maximal ideal $\mathfrak{m}_B$ of B lying over $\mathfrak{m}'$, and elements $a_i', b_i', c_{ij}', k_{ij}' \in B'$ such that

(1) $\kappa(\mathfrak{m}') = \kappa(\mathfrak{m}_B)$ which implies that $\Lambda[x_1,\ldots,x_s]_{\mathfrak{m}'} \subset B_{\mathfrak{m}_B} \subset P$ and P is identified with the completion of B at $\mathfrak{m}_B$, see remark preceding Smoothing Ring Maps, Theorem 07QZ,
(2) $a_i - a_i', b_i - b_i', c_{ij} - c_{ij}', k_{ij} - k_{ij}' \in (\mathfrak{m}')^N P$, and
(3) $f_j(a_1',\ldots,a_u') = \sum c_{ji}' b_i'$ and $\sum k_{ij}' b_j' = 0$.

Set $A = B/(b_1',\ldots,b_r')$ and denote $\mathfrak{m}_A$ the image of $\mathfrak{m}_B$ in A. (Note that A is essentially of finite type over Λ; at the end of the proof we will show how to obtain an A which is of finite type over Λ.) There is a ring map $C \to A$ sending $y_i \mapsto a_i'$ because the a_i' satisfy the desired equations modulo $(b_1',\ldots,b_r')$. Note that $A/\mathfrak{m}_A^N = R/\mathfrak{m}_R^N$ as quotients of $P = B^\wedge$ by property (2) above. Set $x_A = x_C|_{\mathrm{Spec}(A)}$. Since the maps

$$C \to A \to A/\mathfrak{m}_A^N \cong R/\mathfrak{m}_R^N \quad \text{and} \quad C \to R \to R/\mathfrak{m}_R^N$$

are equal we see that x_A and x agree modulo $\mathfrak{m}_R^N$ via the isomorphism $A/\mathfrak{m}_A^N = R/\mathfrak{m}_R^N$. At this point we have shown properties (1) – (5) of the statement of the lemma. To see (6) note that

$$P^{\oplus t} \xrightarrow{K} P^{\oplus r} \xrightarrow{(b_1,\ldots,b_r)} P \quad \text{and} \quad P^{\oplus t} \xrightarrow{K'} P^{\oplus r} \xrightarrow{(b_1',\ldots,b_r')} P$$

are two complexes of P-modules which are congruent modulo $(\mathfrak{m}')^N$ with the first one being exact. By our choice of N above we see from More on Algebra, Lemma 07VF that $R = P/(b_1,\ldots,b_r)$ and $P/(b_1',\ldots,b_r') = B^\wedge/(b_1',\ldots,b_r') = A^\wedge$ have isomorphic associated graded algebras, which is what we wanted to show.

This last paragraph of the proof serves to clean up the issue that A is essentially of finite type over S and not yet of finite type. The construction above gives $A = B/(b_1',\ldots,b_r')$ and $\mathfrak{m}_A \subset A$ with B étale over $\Lambda[x_1,\ldots,x_s]_{\mathfrak{m}'}$. Hence A is of finite type over the Noetherian ring $\Lambda[x_1,\ldots,x_s]_{\mathfrak{m}'}$. Thus we can write $A = (A_0)_{\mathfrak{m}'}$ for some finite type $\Lambda[x_1,\ldots,x_n]$ algebra A_0. Then $A = \mathrm{colim}(A_0)_f$ where $f \in \Lambda[x_1,\ldots,x_n] \setminus \mathfrak{m}'$, see Algebra, Lemma 00CR. Because $p : \mathcal{X} \to (Sch/S)_{fppf}$ is limit preserving on objects, we see that x_A comes from some object $x_{(A_0)_f}$ over $\mathrm{Spec}((A_0)_f)$ for an f as above. After replacing A by $(A_0)_f$ and x_A by $x_{(A_0)_f}$ and $\mathfrak{m}_A$ by $(A_0)_f \cap \mathfrak{m}_A$ the proof is finished. □

References